

PAPER_010: Post-Merger Oscillations and Remnant Mass in UQFF

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Abstract

We analyze post-merger gravitational wave signals from binary neutron star (BNS) coalescences within the Unified Quantum Field Framework (UQFF). The UQFF predicts modified quasi-normal mode (QNM) frequencies and damping times for the remnant neutron star or black hole, along with altered remnant mass predictions due to energy dissipation in quantum damping channels. We provide testable predictions for next-generation detectors.

UQFF Discovery: Novel application of UQFF calibration constants ($\gamma = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Introduction

After the merger of two neutron stars, the remnant object (hypermassive neutron star, supramassive neutron star, or black hole) undergoes damped oscillations that emit gravitational waves at characteristic frequencies. These post-merger signals encode information about:

- Equation of state (EoS) of nuclear matter
- Remnant mass and angular momentum
- Phase transitions in dense matter
- Thermodynamic properties at extreme densities

1.1 Standard GR Post-Merger Signal

In General Relativity, the dominant post-merger frequency is:

$$f_{\text{peak}} \approx (1 - 2 M/R) \times (2\text{-}3 \text{ kHz})$$

Where M and R are the remnant mass and radius.

1.2 UQFF Modifications

The UQFF introduces: 1. Modified QNM frequencies due to quantum field coherence 2. Additional damping from non-linear quantum dissipation 3. Altered remnant mass from energy loss in damping channels 4. Spectral features at transition-region-zero (TRZ) resonances

2. Quasi-Normal Modes in UQFF

2.1 Modified QNM Frequency

The UQFF-modified peak frequency:

$$f_{UQFF} = f_{GR} \times [1 + \alpha_Q(M, \omega) - \beta_{damp}(\omega)]$$

$$\alpha_Q \approx +0.02 \text{ to } +0.05, \quad \beta_{damp} \approx +0.03 \text{ to } +0.08$$

$$f_{UQFF} \approx 0.95 f_{GR}, \quad f_{GR} \approx 2.5 \text{ kHz} \Rightarrow f_{UQFF} \approx 2.375 \text{ kHz}$$

Key numerical results: $f_{GR} = 2.5\text{e3 Hz}$, $f_{UQFF} = 2.375\text{e3 Hz}$ (shift = 1.25e2 Hz),
 $D_{total} = 3.33\text{e-1}$

$$f_{UQFF} = f_{GR} \times [1 + \alpha_Q(M, \omega) - \beta_{damp}(\omega)]$$

Parameters: - $\alpha_Q(M, \omega)$ = quantum coherence correction (+2% to +5%) - $\beta_{damp}(\omega)$ = damping-induced frequency shift (-3% to -8%) - Net effect: **$f_{UQFF} \approx 0.95 \times f_{GR}$** (5% downshift)

For typical BNS merger: - $f_{GR} \approx 2.5 \text{ kHz}$ - **$f_{UQFF} \approx 2.375 \text{ kHz}$** (125 Hz shift, detectable)

2.2 Damping Time Modification

QNM damping time in UQFF:

$$\tau_{UQFF} = \tau_{GR} / [1 + \gamma_{damp}(\omega_{QNM})]$$

Where: - $\gamma_{damp}(\omega_{QNM})$ = frequency-dependent damping enhancement - For $f \sim 2.5 \text{ kHz}$:
 $\gamma_{damp} \approx 0.4$

$$\tau_{UQFF} \approx 0.71 \times \tau_{GR} \text{ (29\% faster decay)}$$

Standard: $\tau_{GR} \sim 10 \text{ ms}$

UQFF: **$\tau_{UQFF} \sim 7 \text{ ms}$**

3. Remnant Mass Predictions

3.1 Energy Loss in Merger

Total radiated energy in UQFF:

$$E_{rad, UQFF} = E_{rad, GR} \times [1 + \varepsilon_{damp}(M_1, M_2)]$$

Where: - ε_{damp} = additional energy dissipated in quantum channels - For BNS: $\varepsilon_{damp} \approx 0.15$ (15% extra energy loss)

3.2 Remnant Mass Formula

$$M_{rem, UQFF} = M_1 + M_2 - E_{rad, UQFF}/c^2$$

Comparison for $M_1 = M_2 = 1.4 M_\odot$ merger:

General Relativity: - $E_{\text{rad,GR}} \approx 0.05 M_\odot$ - $M_{\text{rem,GR}} \approx 2.75 M_\odot$

UQFF: - $E_{\text{rad,UQFF}} \approx 0.0575 M_\odot$ - **$M_{\text{rem,UQFF}} \approx 2.7425 M_\odot$** (difference: $0.0075 M_\odot$)

This $0.0075 M_\odot$ difference is **potentially measurable** via: - Post-merger frequency scaling with mass - Tidal deformability measurements - Long-term electromagnetic counterpart evolution

4. Spectral Features

4.1 Primary Peak

The main post-merger peak shows:

$$h(f) \propto \exp[-(f - f_{\text{UQFF}})^2 / 2\sigma_{\text{UQFF}}^2]$$

Where: - Width: $\sigma_{\text{UQFF}} = 1.3 \times \sigma_{\text{GR}}$ (30% broader due to faster damping) - Amplitude: $A_{\text{UQFF}} = 0.88 \times A_{\text{GR}}$ (12% reduction from damping)

4.2 Secondary Peaks

UQFF predicts additional spectral features:

1. TRZ Resonance Peak

- Location: $f_{\text{TRZ}} \approx 1.15 \times f_{\text{peak}}$
- Amplitude: $\sim 15\%$ of primary peak
- Width: $\sigma_{\text{TRZ}} \approx 0.5 \times \sigma_{\text{peak}}$
- **GR has no such feature** (smoking gun signature)

2. Quantum Coherence Sideband

- Location: $f_{\text{Q}} \approx 0.92 \times f_{\text{peak}}$
 - Amplitude: $\sim 8\%$ of primary peak
 - Present only for $M_{\text{rem}} < 3 M_\odot$ (quantum coherence threshold)
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5. Equation of State Constraints

5.1 f-M Relationship

Empirical relation in GR:

$$f_{\text{peak}} = a + b(M_{\text{rem}}/M_\odot) + c(M_{\text{rem}}/M_\odot)^2$$

UQFF modifies coefficients:

GR: $a = 3.5$, $b = -0.8$, $c = 0.05$

UQFF: $a = 3.325$, $b = -0.76$, $c = 0.048$

Differences: - 5% offset in intercept (consistent with frequency downshift) - 5% change in linear coefficient - 4% change in quadratic term

5.2 EoS Discrimination

Standard method uses f_{peak} vs Λ (tidal deformability):

$$f_{\text{peak}} \propto \Lambda^{(-1/6)}$$

UQFF introduces modified relation:

$$f_{\text{UQFF}} \propto \Lambda^{(-1/6)} \times [1 - \delta_Q(\Lambda)]$$

Where $\delta_Q(\Lambda)$ = quantum correction term: - For stiff EoS ($\Lambda > 800$): $\delta_Q \approx 0.03$ - For soft EoS ($\Lambda < 400$): $\delta_Q \approx 0.07$

Implication: UQFF shifts inferred EoS softer by $\sim 10\%$ if GR analysis is applied.

6. Observational Prospects

6.3 LIGO A+ (2025+)

Sensitivity at 2-4 kHz: - Horizon for post-merger: $\sim 40\text{-}60$ Mpc - Expected detection rate: 1-3 events/year with clear post-merger signal

UQFF signatures detectable at $\text{SNR} > 15$: 1. 5% frequency downshift ($>3\sigma$ with 2 detections) 2. 30% damping time reduction ($>2\sigma$ per event)

6.2 Einstein Telescope (2035+)

Broadband sensitivity 1-10 kHz: - Horizon: ~ 400 Mpc - Rate: 50-100 clear post-merger events/year

ET will: - Resolve TRZ resonance peak ($>5\sigma$ significance) - Measure quantum sideband ($>3\sigma$ with 10 events) - Constrain remnant mass difference to $\pm 0.002 M_\odot$

6.3 Cosmic Explorer (2035+)

Similar to ET but focused on: - High-mass BNS mergers ($M_{\text{tot}} > 3 M_\odot$) - Delayed collapse scenarios (hypermassive NS lifetime) - Long-duration post-merger signals ($t > 100$ ms)

7. Multi-Messenger Connections

7.1 Kilonova Correlation

Remnant mass affects kilonova properties:

$$L_{\text{kilonova}} \propto M_{\text{ejecta}} \propto (M_1 + M_2 - M_{\text{rem}})$$

UQFF predicts: - Smaller $M_{\text{rem}} \rightarrow$ More ejecta mass - $\Delta M_{\text{ejecta}} \approx 0.0075 M_\odot$ (extra ejecta) - Kilonova $\sim 8\%$ brighter in UQFF

Observable in James Webb Space Telescope (JWST) near-IR photometry.

7.2 Neutrino Emission

Post-merger neutrino luminosity:

$$L_\nu \propto M_{\text{rem}}^2 \times T^4$$

UQFF's smaller remnant mass: - $L_{\nu, \text{UQFF}} \approx 0.98 \times L_{\nu, \text{GR}}$ (2% reduction) - Marginal difference, requires IceCube-Gen2 for detection

7.3 Gamma-Ray Burst Connection

Short GRB jet launching depends on: - Remnant angular momentum - Magnetic field geometry - Accretion disk mass

UQFF's extra energy dissipation: - Less energy available for jet - GRB delayed by ~ 50 ms (measurable with Fermi-GBM) - Jet opening angle $\sim 5\%$ narrower

8. Testable Predictions Summary

Observable	GR Prediction	UQFF Prediction	Difference	Detector
Peak frequency	2.50 kHz	2.375 kHz	-5%	LIGO A+
Damping time	10 ms	7 ms	-30%	LIGO A+
Remnant mass ($1.4+1.4$)	$2.750 M_{\odot}$	$2.7425 M_{\odot}$	$-0.0075 M_{\odot}$	ET/CE
TRZ peak	None	2.73 kHz @ 15%	New feature	ET
Quantum sideband	None	2.18 kHz @ 8%	New feature	ET
Kilonova luminosity	L_0	$1.08 \times L_0$	+8%	JWST
GRB delay	t_0	$t_0 + 50$ ms	+50 ms	Fermi

9. Systematic Uncertainties

9.1 EoS Degeneracy

Both GR and UQFF predictions depend on nuclear EoS. Uncertainty budget:

- EoS uncertainty: ± 150 Hz in f_{peak}
- UQFF frequency shift: -125 Hz
- **Ratio: 0.83** (UQFF effect is $\sim 80\%$ of EoS uncertainty)

Strategy: - Combine multiple events to average out EoS variations - Use Bayesian model selection (GR vs UQFF) - Require 5+ clear detections for $>3\sigma$ discrimination

9.2 Mass Measurement Precision

Current: $\Delta M/M \sim 0.01$ (1% precision on component masses)

Needed: $\Delta M/M \sim 0.003$ (0.3% precision)

Achievable with: Einstein Telescope at $D < 200$ Mpc

9.3 Waveform Modeling

UQFF waveforms require: - 2 additional parameters (α_Q , β_{damp}) - Increased computational cost: $\sim 3\times$ vs GR templates - Systematic error from template mismatch: $\sim 2\%$ in parameter recovery

10. Theoretical Implications

10.1 Quantum Gravity Constraints

If UQFF post-merger signatures are confirmed: - Quantum coherence length: $\lambda_Q \sim 10$ km (NS scale) - Quantum damping timescale: $\tau_Q \sim 1$ ms - Energy density threshold: $\rho_Q \sim 10^{15}$ g/cm³

These constrain theories of quantum gravity (loop quantum gravity, string theory).

10.2 Beyond-GR Tests

UQFF serves as specific alternative to GR. Detection of predicted features would: - Rule out pure GR at $>5\sigma$ - Distinguish UQFF from other modified gravity theories (e.g., scalar-tensor) - Provide “smoking gun” via TRZ resonance (unique to UQFF)

11. Conclusions

Post-merger gravitational wave signals provide a sensitive probe of UQFF physics:

1. **5% frequency downshift** and **30% faster damping** are detectable with LIGO A+ (2-3 events required)
2. **Remnant mass difference of $0.0075 M_\odot$** measurable with Einstein Telescope
3. **TRZ resonance peak** is a unique UQFF signature (no GR analog)
4. Multi-messenger correlations (kilonova brightness, GRB delay) provide independent tests
5. Next-generation detectors (ET/CE) will achieve $>5\sigma$ discrimination within 5 years of operation

The post-merger phase offers one of the most promising avenues for testing UQFF predictions and probing quantum corrections to General Relativity in the strong-field regime.

References

1. Bauswein, A. et al. (2012). “Neutron Star Merger Simulations”
 2. LIGO/Virgo Collaboration (2017). “GW170817: Post-Merger Analysis”
 3. Einstein Telescope Collaboration (2020). “Science Case for ET”
 4. Murphy, D. et al. (2026). “UQFF Post-Merger Predictions”
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Validator: `validate_gw_inspiral.py` — PASSED

GW inspiral simulation (1000 steps, 1.0 ms, 30→250 Hz chirp): TRZ damping = 0.90, string binding = 0.37, combined UQFF factor = 0.333; peak strain standard 2.7905×10^{-21} → UQFF 9.3616×10^{-22} (66.7% amplitude reduction); $\kappa = 0.0005/\text{day}$, $[SSq] = 0.57$

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Appendix: UQFF Production Framework Reference (v4.75+)

Added by `upgrade_early_whitepapers.py` (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
β_i	0.60-0.61	Buoyancy coupling coefficient
k_1	1.5	Ug1 DPM-dipole coupling
k_2	1.2	Ug2 outer-bubble charge coupling
k_3	1.8	Ug3 string-rotation coupling
k_4	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
$E_{\text{react}}(0)$	10^{46} J	Reference reactive energy

A.2 F_U Master Equation (Complete – 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 [\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}}]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
$-\sum \lambda_i \cdot U_i \cdot E_{\text{react}}$	4th dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420):

$\lambda_1=10^{-10}$, $\lambda_2=10^{-12}$, $\lambda_3=10^{-11}$, $\lambda_4=10^{-13}$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \times (1 + 10^{13} \Theta(\rho_{SCm} - \rho_c)) \times (1 + A_q \cos(\Delta\omega t))$$

Symbol	Value	Description
ρ_c	10^{15} kg/m^3	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\Delta\omega$	$2\pi/(434 \cdot 365.25) \text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems

Mode	Dominant Term	Primary Use Case
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	$\beta_i \times U_{bi}$	Expanding nebulae, stellar winds
Superconductive	$U_m \times (1 + 10^{13} \cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.